$$\min_{C_1, C_2, \ldots C_k} \left\{ \sum_{i=1}^{k} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{d} \left(x_j^{(i)} - x_j^{(i')} \right)^2 \right\}$$

Greedy algorithm for k-means

1) Randomly assign number from 1 to k for each observation / sample

2) Iterate until the cluster assignments stop changing

a) for each of the k-cluster, compute the cluster centroid. The kth cluster centroid is a vector of d feature means for the observation in k cluster.

b) Assign each observation / sample to the cluster whose centroid is closest (Euclidate distance)

end.

x_1	x_2
1	1
1	2
2	2
8	8
8	9
9	8

$k = 2$

$\boxed{k=2}$ compute centroid

C_1

S_1 (1,1)
S_3 (2,2)
S_4 (8,8)

$\left(\frac{1+2+8}{3}, \frac{1+2+8}{3} \right)$

$(11/3, 11/3)$

C_2

S_2 (1,2)
S_5 (8,9)
S_6 (9,8)

$\left(\frac{1+8+9}{3}, \frac{2+9+8}{3} \right)$

$(19/3, 19/3)$

centroid

$C_1 = (11/3, 11/3)$

$C_2 = (19/3, 19/3)$

Process each point & assign the cluster based on the min distance to the centroid.

x_1

$D(S_1, C_1) = (1,1)\ (3.66, 3.66)$

$= \sqrt{(1-3.66)^2 + (1-3.66)^2}$

$= \sqrt{7.0756 + 7.0756}$

$= \sqrt{14.1512}$

$= 3.761$

$D(S_3, C_1) = (2,2)\ (3.66, 3.66)$

$= \sqrt{(2-3.66)^2 + (2-3.66)^2}$

$= \sqrt{2.7556 + 2.7556}$

$= \sqrt{5.5112}$

$= 2.347$

$D(S_4, C_1) = (8,8)\ (3.66, 3.66)$

$= \sqrt{(8-3.66)^2 + (8-3.66)^2}$

$= \sqrt{(4.34)^2 + (4.34)^2}$

$= \sqrt{37.6712}$

$= 6.137$

$D(S_2, C_1) = (1,2)\ (3.66, 3.66)$

$= 3.135$

$D(S_5, C_1) = (8,9)\ (3.66, 3.66)$

$= 6.881$

$D(S_6, C_1) = (9,8)\ (3.66, 3.66)$

$= 6.881$

$D(S_2, C_2) = (1,2)\ (6.33, 6.33)$

$= \sqrt{(1-6.33)^2 + (1-6.33)^2}$

$= \sqrt{56.8178}$

$= 7.537$

$D(S_5, C_2) = (8,9)\ (6.33, 6.33)$

$= \sqrt{(8-6.33)^2 + (9-6.33)^2}$

$= \sqrt{2.7889 + 7.1289}$

$= \sqrt{9.9178}$

$= 3.149$

$D(S_6, C_2) = (9,8)\ (6.33, 6.33)$

$= 3.149$

$D(S_1, C_2) = (1,1)\ (6.33, 6.33)$

$= 7.537$

$D(S_2, C_2) = (2,2)\ (6.33, 6.33)$

$= 6.123$

$D(S_4, C_2) = (8,8)\ (6.33, 6.33)$

$= 2.361$

C_1		C_2	
S_1	$(1,1)$	S_4	$(8,8)$
S_3	$(2,2)$	S_5	$(8,9)$
S_2	$(1,2)$	S_6	$(9,8)$
	$\left(\frac{1+2+1}{3}, \frac{1+2+2}{3}\right)$		$\frac{25}{3}, \frac{25}{3}$ (8.3)

	$C_1\ (1.3, 1.6)$	$C_2\ (8.3, 8.3)$
S_1	0.67	10.3
S_2	0.50	9.64
S_3	0.80	8.90
S_4	9.26	0.42
S_5	9.9	0.76
S_6	10.0	0.76